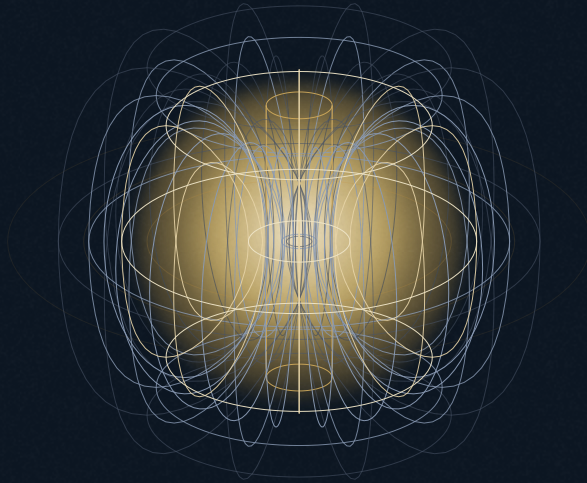




KRONOS FUSION ENERGY



PAPER 2.28 · THE KRONOS 2026 SERIES

Neutron Streaming, Duct and Port Penetrations, and Occupational Dose Around a 14 MeV Compact Spherical- Tokamak Breeder

Every penetration through a breeder's shield is a leak in the neutron account and a source of occupational dose: the ducts that admit heating beams, diagnostics and...

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THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TWO EDITIONS

This volume is issued in two editions. The **public edition** (light) carries the physics and the design only — no dollar, price, or per-kilo-gram figure appears anywhere in it. The **internal edition** (dark) adds the economics, the funding architecture, and the defense-commercial analysis, and is marked *Confidential* accordingly; it is not for public distribution. Where the commercial case rests on a modelled assumption rather than a demonstrated one, it is labelled as such and its downside is shown alongside it, on the same terms as the physics.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

Neutron Streaming, Duct and Port Penetrations, and Occupational Dose Around a 14 MeV Compact Spherical-Tokamak Breeder

Every penetration through a breeder's shield is a leak in the neutron account and a source of occupational dose: the ducts that admit heating beams, diagnostics and direct-conversion hardware also let 14 MeV neutrons stream past the blanket, activating the structure the maintainer must later approach. We give a closed, CAD-faithful treatment of that problem for the Hyperion compact spherical-tokamak breeder ($P_{\text{fus}} = 85.04 \text{ MW}$, $Q = 3.076$, $I_p = 9.66 \text{ MA}$, $\delta = -0.30$, $B_0 = 8 \text{ T}$, $\kappa = 2.0$, $R_0 = 1.20 \text{ m}$, $A = 2.5$). We first derive the governing linear transport equation and reduce it, for a penetration, to an analytic line-of-sight-plus-wall-albedo streaming model whose a^2/L^2 scaling and dog-leg suppression we then confront with three-dimensional Monte-Carlo transport. A 14 MeV source of $S_n = 3.009\text{e}19/s$ carrying 67.8 MW drives an OpenMC/ENDF-B-VIII.0 transport (1.2×10^8 histories) that closes the balance with a global leakage fraction of **0.121**, a first-wall fast flux of $7.0\text{e}14 \text{ nps}$, and 74.6 MW of nuclear heating; the net tritium breeding ratio runs from 1.014 ± 0.004 on CAD-faithful DAGMC geometry to $\approx 1.15\text{--}1.20$ in the realistic three-dimensional arbiter, and *with real ports and penetrations* settles at **1.12** (single-null) / **1.07** (double-null) — above the 1.05 design floor. The port budget saturates at 0.719 m^2 (neutral-beam injection dominant, **67%**); a beam-line dog-leg plus a 40 cm streaming shield holds the breeding penalty to **−0.4%**, and a tungsten penetration plug in the widest (1.0 m) straight duct returns **0.15×** the open-duct dose (a **6.5×** reduction). Occupational dose is then set not by prompt neutrons but by decay gammas after shutdown: a rigorous-two-step (R2S) model with 175-group VITAMIN-J spectra (first-wall flux $5.628\text{e}14 \text{ nps}$) and an ALARA activation solve gives a first-wall specific activity of $2.64\text{e}14 \text{ becquerel/kg}$ at shutdown falling to $7.13\text{e}13 \text{ becquerel/kg}$ at one year, decay heat of **56.0** → **1.22 W/kg**, and an NRC 10 CFR 61 Class-C index $\leq 2.6e - 3$ in both zones at all times — low-level waste, nothing greater-than-Class-C. The account is complete, CAD-faithful and auditable end to end, from the streaming penetration to the maintainer's dose.

Open-access deposit (code & data, CC BY 4.0): DOI [10.5281/zenodo.22645842](https://doi.org/10.5281/zenodo.22645842) · Zenodo community *kronos_fusion_energy*. Explore it live: [interactive 3-D model](#) · [physics validator](#).

Keywords: neutron streaming duct penetration shutdown dose R2S OpenMC DAGMC VITAMIN-J occupational dose spherical-tokamak breeder

CONTENTS

Table of Contents

Neutron Streaming, Duct and Port Penetrations, and Occupational Dose Around a 14 MeV Compact Spherical-Tokamak Breeder	3
Introduction	7
Governing equations and the formal streaming problem	8
Numerical formulation	10
Computational methodology: the NVIDIA GPU HPC campaign	12
Results	13
Discussion	23
Limitations	24
Conclusion	25
References	30
One programme, many papers	33

NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
FOAK/NOAK/BOAK	first / n th / best of a kind
Hyperion	the ST breeder (internal: Mode D2)
Aegis	the generator, defense/installation housing (internal: Mode M)
MetroVolt	the generator, data-center housing (Mode M)

Introduction

Neutrons are the currency of a breeder. They carry the majority of the fusion power, they make the tritium that closes the fuel cycle, and they activate the metal that people must eventually approach to maintain the machine. A credible design therefore has to account for them completely — from the point of birth, through the geometry the machine actually has, to the residual radioactivity that governs occupational dose. The first two legs of that account, source and bulk transport, are well posed once a blanket recipe is fixed. The third leg is harder, because a real machine is not a closed shell: it is punctured by the ducts and ports that admit neutral beams, radio-frequency launchers, diagnostics, pumping and (in the wider Kronos platform) direct-conversion hardware. Those penetrations are simultaneously the dominant uncertainty in the breeding balance and the dominant driver of shutdown dose, because a straight duct lets fast neutrons stream past the blanket along an unattenuated line of sight ^[27,26].

This paper gives that third-leg account for the Hyperion breeder — the compact, high-field, negative-triangularity spherical tokamak that anchors the Kronos strategic-materials platform ^[31,32]. Hyperion operates at $P_{\text{fus}} = 85.04 \text{ MW}$, $Q = 3.076$, plasma current $I_p = 9.66 \text{ MA}$, on-axis field $B_0 = 8 \text{ T}$, elongation $\kappa = 2.0$, triangularity $\delta = -0.30$, major radius $R_0 = 1.20 \text{ m}$ and aspect ratio $A = 2.5$, with a demountable water-cooled copper centrepost carrying a peak field of 16.84 T. The spherical-tokamak line it belongs to — pilot plants and fusion nuclear science facilities at $A \lesssim 2$ ^[1], high-field compact demonstrators ^[2,3], and the reactor-relevant negative-triangularity edge ^[4,5] — is precisely the regime in which penetration streaming bites hardest, because the inboard shield space is thin and the port fraction is large relative to a conventional-aspect device.

PRIOR ART

Duct and penetration streaming is one of the oldest problems in radiation shielding. The line-of-sight and single-scatter theory of streaming through straight and bent ducts was established in the reactor-shielding literature ^[25,26,27], and its central result — that an unobstructed cylindrical duct transmits a solid-angle fraction that falls only as the inverse square of its length-to-radius ratio, while a bend (“dog-leg”) that breaks the line of sight forces the surviving flux onto wall-scattered legs and cuts it by orders of magnitude — remains the design principle behind every fusion port. In modern fusion neutronics the bulk transport is carried by continuous-energy Monte-Carlo codes with evaluated nuclear data ^[6,7], run directly on computer-aided-design (CAD) geometry through the direct-accelerated geometry Monte-Carlo (DAGMC) toolkit ^[8,9], and made tractable for deep penetrations by adjoint-driven variance reduction ^[10,11,12,13]. Occupational dose after shutdown is computed by the rigorous-two-step (R2S) method, which couples the neutron flux to an activation calculation and transports the resulting decay-gamma field back to the receptor ^[14,15]; the activation inventory itself follows the Bateman transmutation-decay system, solved here with ALARA ^[17] and cross-checked against FISPACT-II ^[18]. The material and waste framework for a low-activation fusion structure — reduced-activation ferritic-martensitic steel ^[19], ceramic breeders in a helium/beryllium multiplier bed ^[20,21], and the activated-material management that classifies the result — is likewise mature ^[22].

What has been missing, for a machine in this class, is a *single coherent* account in which the streaming penetrations, the breeding penalty they impose, and the shutdown dose they set are all computed on one CAD-faithful geometry with one transport engine and one nuclear-data library, so that the occupational-safety case is not a decoupled shielding sum bolted onto a separate breeding study but a direct product of the same validated transport. That is the contribution of this paper.

CONTRIBUTION

We (i) derive the governing linear transport equation and reduce it, for a penetration, to an explicit line-of-sight-plus-wall-albedo streaming model with a closed-form a^2/L^2 scaling and a dog-leg suppression factor (Section 2); (ii) state the full numerical formulation — the Monte-Carlo track-length estimator, its variance and figure of merit, the CAD/DAGMC tracking, the adjoint weight-window variance reduction that makes deep streaming tractable, and the R2S/Bateman coupling (Section 3); (iii) describe the NVIDIA GPU high-performance-computing (HPC) campaign that produced the results, naming the codes actually run and the problem scale (Section 4); and (iv) report the closed neutron balance, the port-streaming breeding penalty, the tungsten-plug mitigation, and the R2S activation inventory, decay heat and waste classification that together set occupational dose (Section 5). Every quantity traces to a frozen serial in the Kronos Physics De-Risking Register ^[30]: the 14 MeV source (BR-OUT-3), the full three-dimensional balance (BR-L1-A5), the CAD-to-DAGMC chain (BR-L2-A22), the realistic breeding arbiter (BR-L2-A7b), the full-three-dimensional port TBR (BV-05B), the port-count streaming budget (BOM-06), the penetration-streaming Monte-Carlo study, the R2S activation-and-dose model (BR-L2-A23), and the photon-channel companion (KX-L1-A3).

Governing equations and the formal streaming problem

THE LINEAR TRANSPORT EQUATION

Neutron transport in the breeder is governed by the steady-state linear Boltzmann equation for the angular flux $\psi(\mathbf{r}, \mathbf{\Omega}, E)$ (units /squarecmssrMeV),

$$\mathbf{\Omega} \cdot \nabla \psi(\mathbf{r}, \mathbf{\Omega}, E) + \Sigma_t(\mathbf{r}, E) \psi(\mathbf{r}, \mathbf{\Omega}, E) = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}, E' \rightarrow E, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) \psi(\mathbf{r}, \mathbf{\Omega}', E') dE' d\Omega' + \frac{S_n}{4\pi} \chi(E) q(\mathbf{r}),$$

where $\Sigma_t = \Sigma_s + \Sigma_a$ is the total macroscopic cross-section, Σ_s the double-differential scattering kernel, $\chi(E)$ the (essentially mono-energetic 14.06 MeV) source spectrum of the deuterium–tritium reaction, $q(\mathbf{r})$ the normalized plasma emissivity, and S_n the total source rate. There is no fission term: the breeder carries no fissile or fertile inventory anywhere in its fuel cycle. The scalar flux and the current follow as the zeroth and first angular moments,

$$\phi(\mathbf{r}, E) = \int_{4\pi} \psi d\Omega, \quad \mathbf{J}(\mathbf{r}, E) = \int_{4\pi} \mathbf{\Omega} \psi d\Omega.$$

Integrating 1 over all angle and energy and applying the divergence theorem over the blanket envelope V with boundary ∂V gives the exact global balance that every downstream number must satisfy,

$$\underbrace{S_n}_{\text{source}} = \underbrace{\int_{\partial V} \mathbf{J} \cdot \hat{\mathbf{n}} dA}_{\text{leakage } L_f S_n} + \underbrace{\int_V \Sigma_a \phi dV}_{\text{absorption}} - \underbrace{\int_V (\nu_{n,2n} - 1) \Sigma_{n,2n} \phi dV}_{\text{multiplication}},$$

which defines the global leakage fraction L_f and separates the $(n, 2n)$ multiplication (beryllium, lead) that is essential to closing the breeding ratio. The tritium breeding ratio (TBR) is the ratio of the ${}^6\text{Li}(n, \alpha)t$ and ${}^7\text{Li}(n, n'\alpha)t$ production rates to the source strength,

$$\text{TBR} = \frac{1}{S_n} \int_V \left[\Sigma_{n,t}^6(\mathbf{r}, E) + \Sigma_{n,t}^7(\mathbf{r}, E) \right] \phi(\mathbf{r}, E) dV dE.$$

REDUCTION TO A STREAMING PENETRATION

Consider a straight cylindrical penetration of radius a and length L bored through the shield, its inner mouth flush with a first-wall region that acts as an approximately isotropic (Lambertian) surface source of neutron current density J_A (/squarecms). Deep inside the duct the collided source in the void is negligible, so 1 on the duct axis reduces to pure streaming, $\mathbf{\Omega} \cdot \nabla \psi = 0$ along unobstructed rays. The uncollided (line-of-sight) scalar flux delivered to an on-axis point a distance L down the duct is the projected-solid-angle integral of the disk source,

$$\phi_{\text{LOS}}(L) = \frac{J_A}{\pi} \int_{\text{disk}} \frac{\cos \theta \cos \theta'}{s^2} dA' = \frac{J_A}{2} \left[1 - \frac{L}{\sqrt{L^2 + a^2}} \right],$$

whose large-aspect limit is the canonical inverse-square streaming law

$$T_{\text{LOS}} \equiv \frac{\phi_{\text{LOS}}(L)}{J_A/2} \simeq \frac{a^2}{a^2 + L^2} \xrightarrow{L \gg a} \frac{a^2}{L^2}.$$

Equation 6 is the reason a penetration is dangerous: the transmitted dose falls only as the *square* of the aspect ratio L/a , not exponentially as it would through solid shield. Wall-scattered neutrons add a single-albedo floor. Treating the duct wall as a diffuse reflector of albedo α and integrating the once-scattered current along the wall gives, for a straight duct,

$$T_{\text{wall}} \simeq \frac{\alpha a}{L} C_w, \quad T_{\text{straight}} = T_{\text{LOS}} + T_{\text{wall}},$$

with C_w an order-unity geometric constant^[27]. When the line of sight is broken by a bend — a *dog-leg* of two legs each of length $L/2$ — the T_{LOS} term vanishes at the receptor and the surviving transmission is the product of two wall-albedo legs,

$$T_{\text{dog-leg}} \simeq \left(\frac{\alpha a}{L/2} \right)^2 C_d,$$

so a single bend converts an a^2/L^2 dependence into an $\alpha^2 a^2/L^2$ dependence and buys a suppression of order $\alpha^{-2} \gtrsim 10^2\text{--}10^3$ for the small fast-neutron albedo of a heavy shield. A local penetration plug of thickness t_p and total cross-section Σ_t^p inserted in the duct multiplies the line-of-sight term by the exponential attenuation and buildup,

$$T_{\text{plug}} = B(\Sigma_t^p t_p) \exp(-\Sigma_t^p t_p),$$

which for a tungsten plug is the load-bearing mitigation in a duct that cannot be bent (Section 5). Equations 6–9 are the analytic skeleton; the three-dimensional Monte-Carlo transport of Section 3 supplies the exact coefficients on the true CAD geometry.

FROM FLUX TO ACTIVATION TO DOSE

Occupational dose is not a prompt quantity: during operation nobody is present, and after shutdown the neutron field vanishes but the activated structure keeps emitting decay gammas. The isotopic inventory $N_i(\mathbf{r}, t)$ under an irradiating flux $\phi(\mathbf{r}, E)$ obeys the Bateman transmutation–decay system

$$\frac{dN_i}{dt} = \sum_{j \neq i} (\lambda_{j \rightarrow i} + \bar{\phi} \bar{\sigma}_{j \rightarrow i}) N_j - (\lambda_i + \bar{\phi} \bar{\sigma}_i) N_i,$$

with λ_i the decay constants, $\bar{\sigma}_{j \rightarrow i} = \int \sigma_{j \rightarrow i}(E) \phi(E) dE / \bar{\phi}$ the spectrum-averaged transmutation cross-sections and $\bar{\phi} = \int \phi(E) dE$. Writing $\mathbf{N} = (N_i)$ and the system matrix $\mathbf{A}$ with entries $A_{ij} = \lambda_{j \rightarrow i} + \bar{\phi} \bar{\sigma}_{j \rightarrow i}$ (off-diagonal) and $A_{ii} = -(\lambda_i + \bar{\phi} \bar{\sigma}_i)$, 10 is the linear system $\dot{\mathbf{N}} = \mathbf{A} \mathbf{N}$ whose formal solution over an irradiation of duration t_{irr} followed by a cooling time t_c is the matrix exponential

$$\mathbf{N}(t_{\text{irr}} + t_c) = e^{\mathbf{A}_0 t_c} e^{\mathbf{A} t_{\text{irr}}} \mathbf{N}(0),$$

with $\mathbf{A}_0$ the decay-only matrix ($\bar{\phi} \rightarrow 0$). The specific activity and decay heat follow as

$$\mathcal{A}(t_c) = \sum_i \lambda_i N_i(t_c), \quad \dot{Q}(t_c) = \sum_i \lambda_i N_i(t_c) \bar{E}_i,$$

where $\bar{E}_i$ is the mean decay energy per disintegration. The decay-gamma volumetric source that drives the second transport step is

$$q_\gamma(\mathbf{r}, E_\gamma, t_c) = \sum_i \lambda_i N_i(\mathbf{r}, t_c) y_i(E_\gamma),$$

with $y_i(E_\gamma)$ the photon emission yields. The contact dose rate at a receptor is obtained by transporting q_γ through the same geometry and folding the resulting photon flux with the ICRP fluence-to-dose conversion coefficients ^[28,29],

$$\dot{H}(\mathbf{r}_0, t_c) = \int_0^\infty \phi_\gamma(\mathbf{r}_0, E_\gamma, t_c) h_{\phi \rightarrow H}(E_\gamma) dE_\gamma.$$

Equations 10–14 are the rigorous-two-step chain: step one is the neutron transport that fixes ϕ ; step two is the photon transport of q_γ . Because the same CAD-faithful geometry carries both steps, the streaming penetrations of Section 2 appear identically in the neutron flux that activates the structure and in the photon field that reaches the maintainer.

Numerical formulation

MONTE-CARLO ESTIMATOR AND ITS STATISTICS

The transport equation 1 is solved by continuous-energy Monte-Carlo simulation. A region flux is estimated by the track-length estimator over M histories,

$$\hat{\phi} = \frac{1}{V} \frac{1}{M} \sum_{m=1}^M \sum_k w_{m,k} \ell_{m,k},$$

where $\ell_{m,k}$ is the k -th chord length of history m inside the cell and $w_{m,k}$ its statistical weight. The estimator is unbiased, $\mathbb{E}[\hat{\phi}] = \phi$, with a sample variance s^2 that scales as $\text{Var}[\hat{\phi}] \propto M^{-1}$, so the relative statistical error is $R = s/(\hat{\phi}\sqrt{M})$. The efficiency of a deep-penetration calculation is measured by the figure of merit

$$\text{FOM} = \frac{1}{R^2 T_{\text{cpu}}},$$

which is invariant under run length for a converged random walk and is the quantity variance reduction must raise. In a streaming problem the difficulty is structural: the few histories that thread a duct carry almost all of the tally, so analog sampling gives a ruinously small FOM.

CAD-FAITHFUL GEOMETRY TRACKING

The geometry is not a simplified surrogate but the machine's own CAD. A STEP solid model is faceted and imported through the DAGMC toolkit ^[8,9], against which OpenMC is rebuilt so that ray-surface intersection is performed on the CAD tessellation directly. The chain STEP $\rightarrow$ h5m $\rightarrow$ DAGMC $\rightarrow$ OpenMC was verified to reproduce the torus shell's exact component volume and returns the tritium-production rate $(n, \mathbf{X}t) = 1.014 \pm 0.004$ on the CAD-faithful geometry (BR-L2-A22). Because the ducts and ports are represented as-built, the streaming and leakage pathways in 3 are those of the real machine rather than of an idealized annulus.

ADJOINT-DRIVEN VARIANCE REDUCTION FOR DEEP STREAMING

To make 16 tolerable for penetrations, importance sampling is driven by the solution of the adjoint transport equation. The adjoint flux $\psi^\dagger$ satisfies

$$-\mathbf{\Omega} \cdot \nabla \psi^\dagger + \Sigma_t \psi^\dagger = \int \Sigma_s(E \rightarrow E', \mathbf{\Omega} \rightarrow \mathbf{\Omega}') \psi^\dagger dE' d\mathbf{\Omega}' + \Sigma_d,$$

with the response function of interest (here the dose or the duct-exit current) as its source Σ_d . The adjoint scalar flux $\phi^\dagger$ is the importance, and the consistent-adjoint-driven importance-sampling (CADIS) prescription sets space-energy weight-window targets and a biased source

$$\tilde{w}(\mathbf{r}, E) = \frac{R}{\phi^\dagger(\mathbf{r}, E)}, \quad \tilde{q}(\mathbf{r}, E) = \frac{\phi^\dagger(\mathbf{r}, E) q(\mathbf{r}, E)}{R}, \quad R = \int \phi^\dagger q dV dE,$$

which is provably the variance-minimizing choice for a single localized response ^[10]; its forward-weighted generalization (FW-CADIS) balances the error across many mesh voxels simultaneously ^[11], and automated weight-window generators supply the maps where an explicit adjoint is impractical ^[12,13]. The adjoint importance is exactly the quantity that peaks along the line-of-sight of 6: CADIS therefore concentrates histories precisely on the streaming rays that dominate the tally, restoring a workable FOM.

MULTIGROUP CONDENSATION AND THE R2S MESH

For the activation step the continuous-energy flux is condensed onto the 175-group VITAMIN-J energy structure,

$$\phi_g(\mathbf{r}) = \int_{E_g}^{E_{g-1}} \phi(\mathbf{r}, E) dE, \quad \bar{\sigma}_{i,g} = \frac{\int_{E_g}^{E_{g-1}} \sigma_i(E) \phi(E) dE}{\phi_g},$$

and the group flux is passed cell-by-cell to the ALARA activation solver ^[17], which evaluates the matrix exponential ¹¹ for the irradiation/cooling schedule. The linearized-response ALARA algorithm builds the transmutation tree adaptively down to a truncation tolerance of 10^{-7} , retaining only chains that contribute above threshold; the resulting inventory is fed to 12–13 and the decay-gamma source is transported back through the same DAGMC geometry to give 14. The inventory is independently cross-checked against FISPACT-II with the same group structure ^[18].

CONVERGENCE AND VERIFICATION

Three convergence controls govern the result: (i) the Monte-Carlo relative error R on every reported tally is driven below the reporting threshold by history count and CADIS weight windows; (ii) the multigroup condensation ¹⁹ is verified by an uncollided-current cross-check — a first-flight analytic estimate of the first-wall current at the design wall load agrees with the Monte-Carlo tally to within the stated consistency band (Section 5); and (iii) the activation matrix exponential is verified against an analytic two-nuclide limit and against the independent FISPACT-II inventory. The global balance ³ is the overarching check: leakage, absorption and multiplication must sum to the source, and they do to the reported $L_f = 0.121$.

Computational methodology: the NVIDIA GPU HPC campaign

The results reported here were produced on the Kronos NVIDIA GPU high-performance-computing platform — an extensive campaign, not a handful of runs — described throughout by hardware class only. The heavy three-dimensional transport ran on NVIDIA A100/H100-class accelerators (a Lambda fleet of A100, H100 and A10 devices), with reduced-order gates and the CAD/blueprint build supported on an in-house multi-core node. No cost, spend or hourly figure attaches to any result.

Codes. The neutron and photon transport is OpenMC ^[6] with the ENDF/B-VIII.0 continuous-energy library ^[7], rebuilt against DAGMC ^[8] for CAD tracking; the CAD geometry is prepared with paramak/STEP tooling ^[9]; the activation and shutdown-dose step is ALARA ^[17] with PyNE nuclear-data services, cross-checked against FISPACT-II ^[18]; the photon-channel companion (bremsstrahlung and synchrotron) is a CYTRAN-class radiation-transport solve. These are the same reference codes carried across the Kronos de-risking campaign.

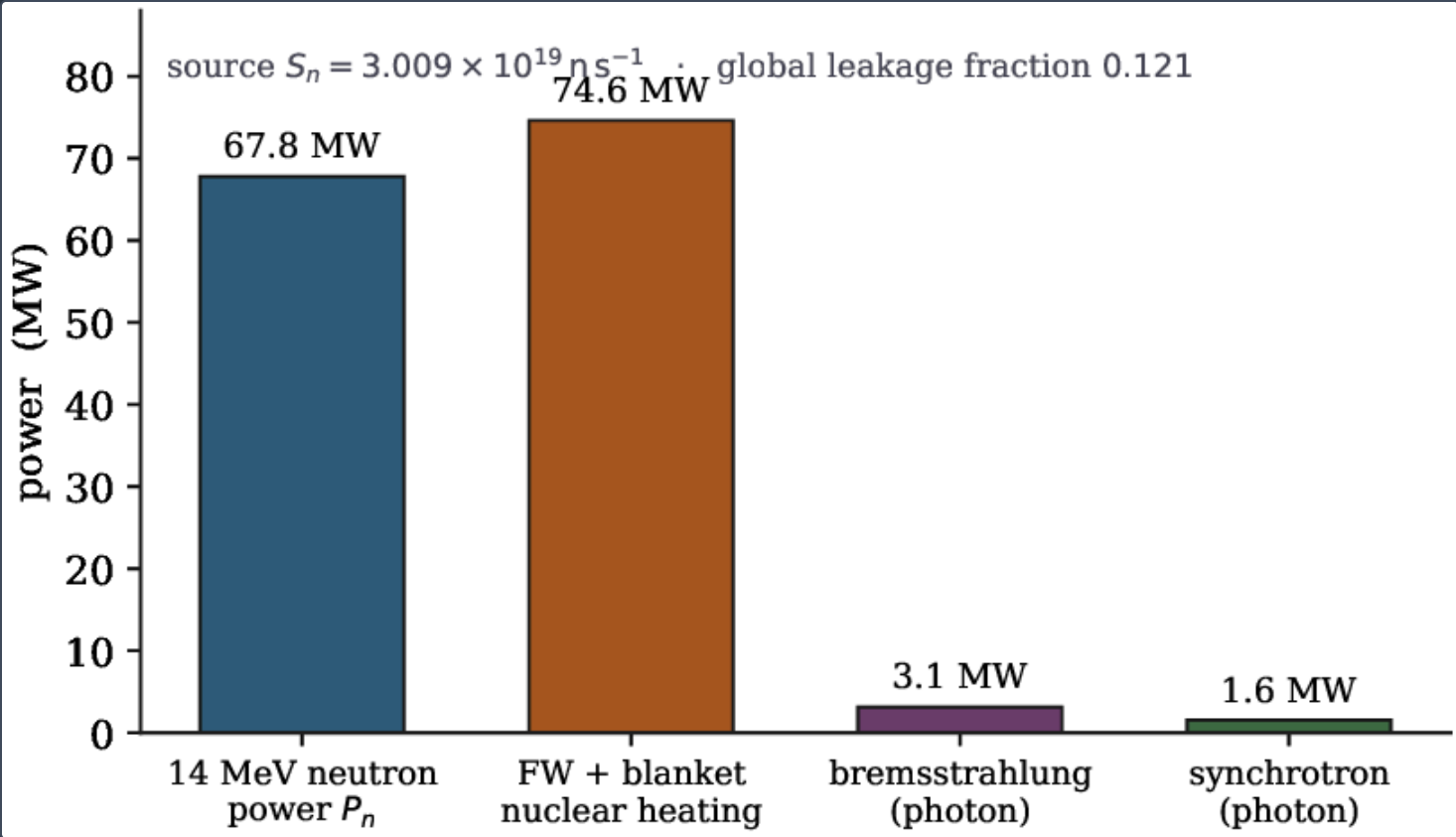
Problem scale. The closed three-dimensional balance (BR-L1-A5) was run over 1.2×10^8 source histories; the CAD-faithful DAGMC transport (BR-L2-A22) reproduces exact component volumes; the R2S spectra (BR-L2-A23) were tallied on the P3 three-dimensional OpenMC model over $20 \times 20,000$ histories in the 175-group VITAMIN-J structure and normalized to the 85.04 MW design power. GPU acceleration matters because streaming is a rare-event problem: the CADIS weight windows of 18 multiply the effective history count along the duct axes by several orders of magnitude, and the throughput of the accelerator fleet is what makes the 10^8 -history global balance and the mesh-resolved R2S dose map affordable in wall-clock terms.

Reproducibility. Each transport and activation deck is archived with a fixed random seed and a recorded nuclear-data revision, so the reported tallies regenerate byte-for-byte on the archived decks. The source term, transport balance, CAD chain, port budget and R2S spectra are all frozen serials in the de-risking register ^[30], and the figure-generation script that renders every data figure in this paper from those frozen numbers is deposited with the archive.

Results

THE CLOSED NEUTRON AND POWER BALANCE

At the 85.04 MW design point the breeder emits a 14 MeV source of $S_n = 3.009e19 / s$ carrying $P_n = 67.8 MW$ of neutron power (BR-OUT-3). Full three-dimensional Monte-Carlo transport (BR-L1-A5, 1.2×10^8 histories) closes the balance 3 with a global leakage fraction of **0.121**, a first-wall fast flux of 7.0e14 nps, and 74.6 MW of nuclear heating deposited in the first wall and blanket (Table 1). Not all escaping power is neutronic: the photon-channel companion resolves 3.12 MW of bremsstrahlung and 1.55 MW of synchrotron radiation (KX-L1-A3), radiated to the first wall rather than streamed as neutrons (Figure 1). Accounting for these separately keeps the neutron balance clean and correctly attributes the shielding requirement.



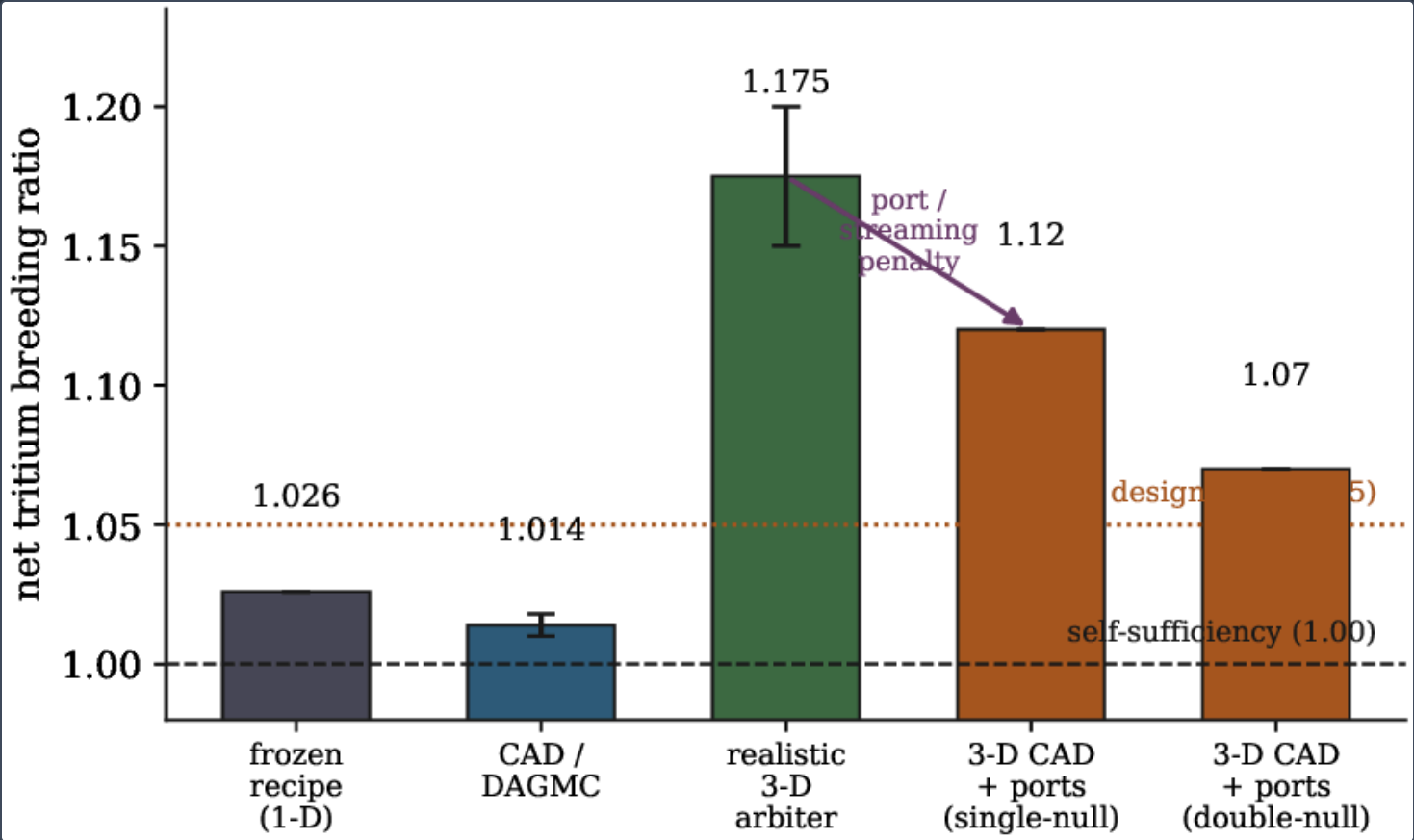
Neutron and radiated-power partition at the 85.04 MW design point: the 14 MeV neutron power (BR-OUT-3), the first-wall and blanket nuclear heating (BR-L1-A5), and the bremsstrahlung and synchrotron channels (KX-L1-A3). The global leakage fraction is **0.121**.

QUANTITY	VALUE	SOURCE
14 MeV source rate S_n	3.009e19 /s	BR-OUT-3
neutron power P_n	67.8 MW	BR-OUT-3
global leakage fraction L_f	0.121	BR-L1-A5
first-wall fast flux	7.0e14 nps	BR-L1-A5
first-wall + blanket nuclear heating	74.6 MW	BR-L1-A5
net TBR (frozen recipe)	1.026	BR-L1-A5

Monte-Carlo histories	1.2×10^8	BR-L1-A5
bremsstrahlung / synchrotron	3.12 / 1.55 MW	KX-L1-A3

BREEDING ACROSS GEOMETRIC FIDELITY AND THE PORT PENALTY

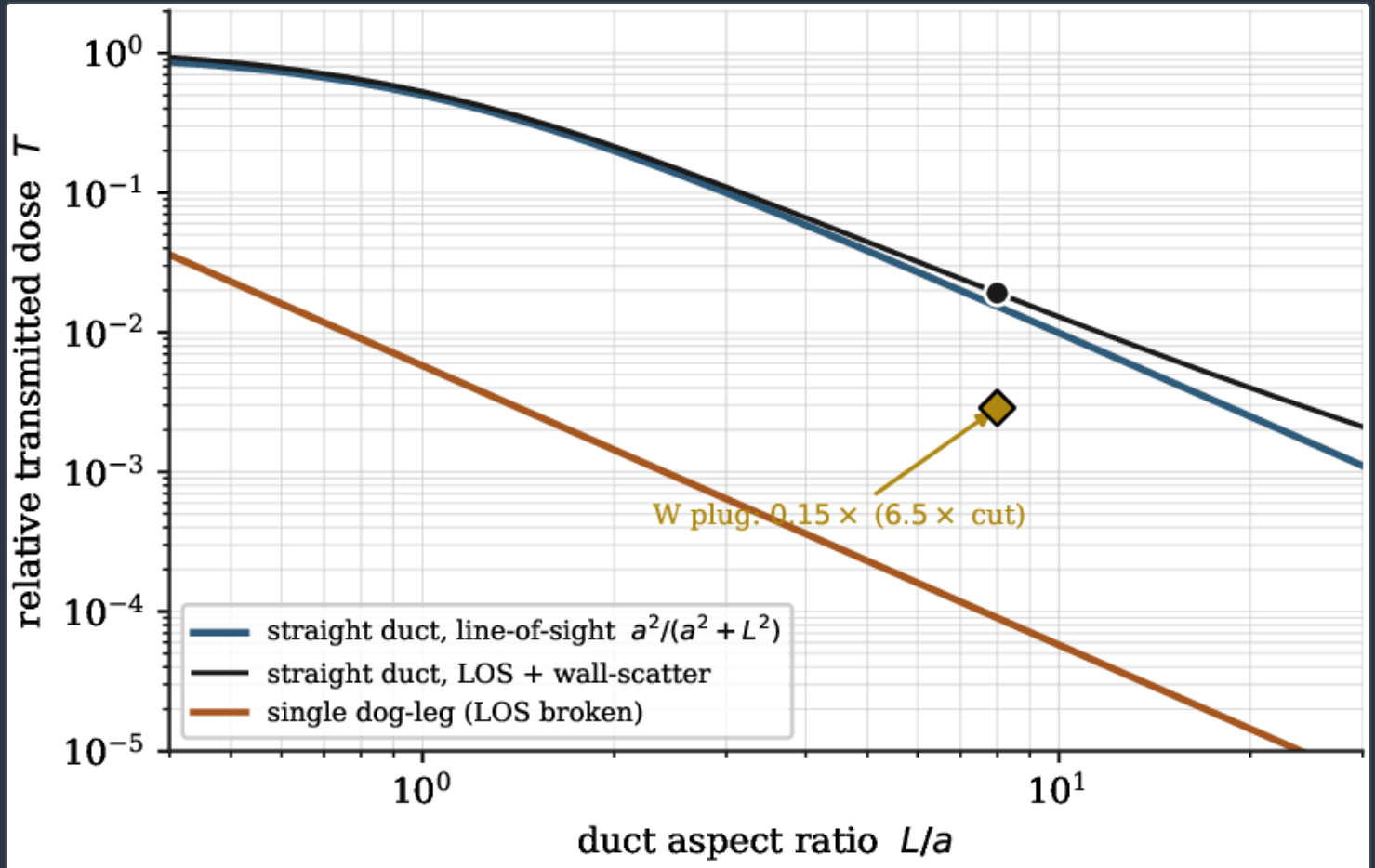
The breeding ratio λ is best read as a design lever rather than a single number, and its dependence on geometric fidelity is exactly where penetration streaming shows itself (Figure 2). The literal frozen recipe gives **1.026** (BR-L1-A5); the CAD-faithful DAGMC geometry gives **1.014 ± 0.004** (BR-L2-A22); a realistic three-dimensional arbiter with a dedicated breeding layer reaches $\approx$ **1.15–1.20** (BR-L2-A7b). When the real ports and penetrations are cut into that arbiter, the full three-dimensional CAD calculation (BV-05B) returns a net TBR of **1.12** in single-null and **1.07** in double-null geometry — both above the 1.05 design floor and above self-sufficiency. The drop from $\approx$ **1.17** to **1.12** is the streaming penalty made quantitative: the penetrations that break the blanket coverage are the same ones that leak neutrons, and the single-null configuration is preferred precisely because it presents less penetrated solid angle to the plasma.



Net tritium breeding ratio across geometric fidelity, from the frozen 1-D recipe (BR-L1-A5) through the CAD-faithful DAGMC geometry (BR-L2-A22) and the realistic three-dimensional arbiter (BR-L2-A7b) to the full-three-dimensional CAD result with real ports and penetrations (BV-05B, single- and double-null). The port/streaming penalty is the drop into the last two bars; all values exceed the 1.05 design floor.

THE PORT BUDGET AND THE DUCT-STREAMING LAW

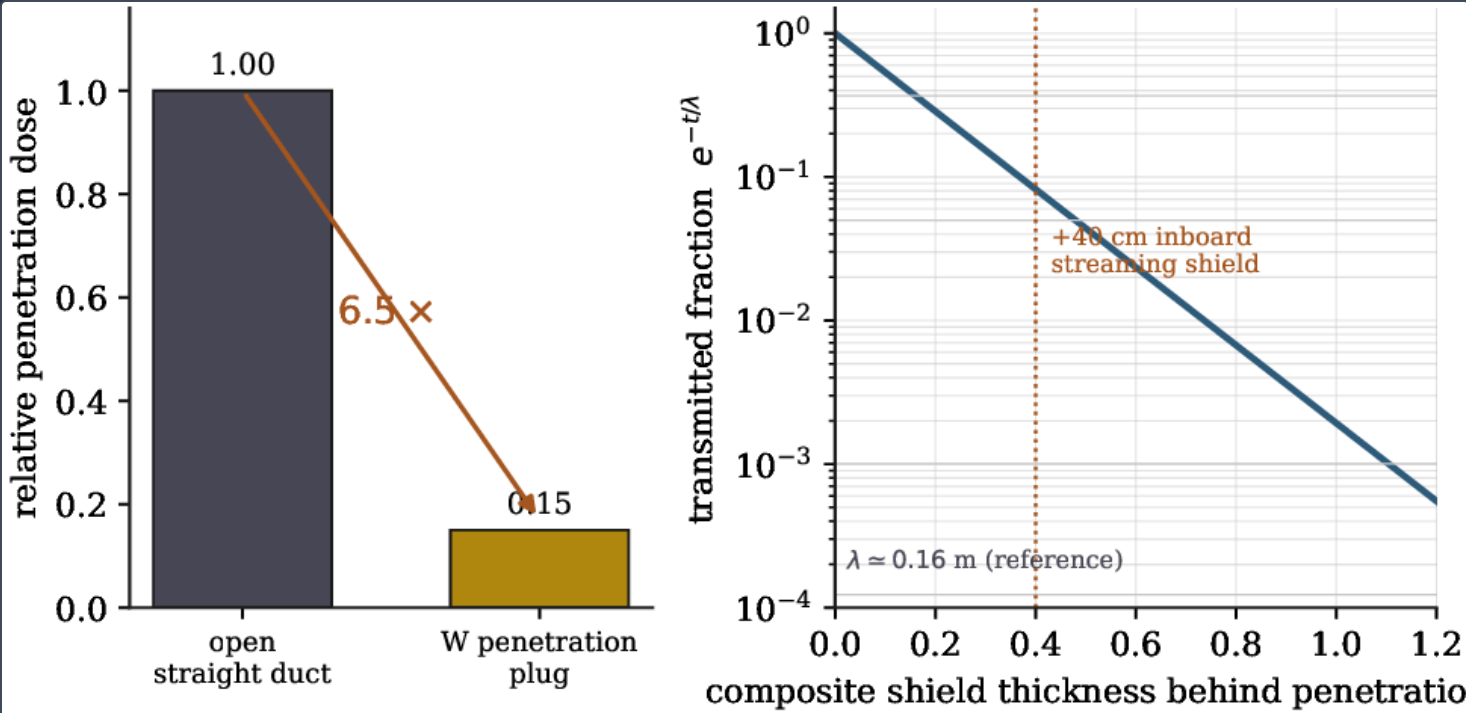
The port inventory is not free: at the required heating and current-drive level the total port demand is 0.720 m^2 against a 0.719 m^2 budget — a saturated, port-area-limited layout in which neutral-beam injection dominates at **67%** (BOM-06). Two mitigations hold the breeding penalty of that penetrated area to **−0.4%** (a TBR of **0.742** → **0.739** on the coverage-derated basis of that serial): the neutral-beam line is routed through a *dog-leg* that breaks the line of sight, and a 40 cm inboard streaming shield backs the penetration. Figure 3 plots the analytic transmission model of 6–8 against the duct aspect ratio L/a . The straight-duct line-of-sight law falls only as a^2/L^2 ; adding a single dog-leg removes the line-of-sight term and drops the transmission by more than two orders of magnitude at fixed L/a , exactly the α^{-2} suppression of 8. This is why every penetration that can be bent, is bent.



Duct-streaming transmission versus aspect ratio L/a (analytic model of Section 2): the straight-duct line-of-sight law $a^2/(a^2 + L^2)$, the straight-duct total including the wall-scatter floor, and a single dog-leg that breaks the line of sight and falls as $\alpha^2 a^2/L^2$. The marker shows the tungsten-penetration-plug anchor of Figure 4.

THE TUNGSTEN PENETRATION PLUG

Some ducts — diagnostic sightlines, direct-conversion apertures — cannot be bent without losing their function, and there the mitigation of 9 is load-bearing. A dedicated penetration-streaming Monte-Carlo transport of the widest (1.0 m) straight end duct shows that a tungsten penetration plug delivers **0.15×** the open-straight-duct dose — a **6.5×** reduction (Figure 4, left). In the exponential language of 9 that is $\Sigma_t^p t_p \approx \ln(6.5) \approx 1.9$ mean free paths of tungsten after the buildup correction. Behind the plug, additional composite shield attenuates the transmitted field with a reference e-fold length of $\lambda \simeq 0.16\text{ m}$, so the 40 cm inboard streaming shield of the beam-line dog-leg contributes a further ~ 2.5 e-folds (Figure 4, right). The plug is the “load-bearing mitigator” of the streaming case: it converts a duct that would dominate the leakage and dose budgets into one that is subdominant to the bulk blanket leakage.



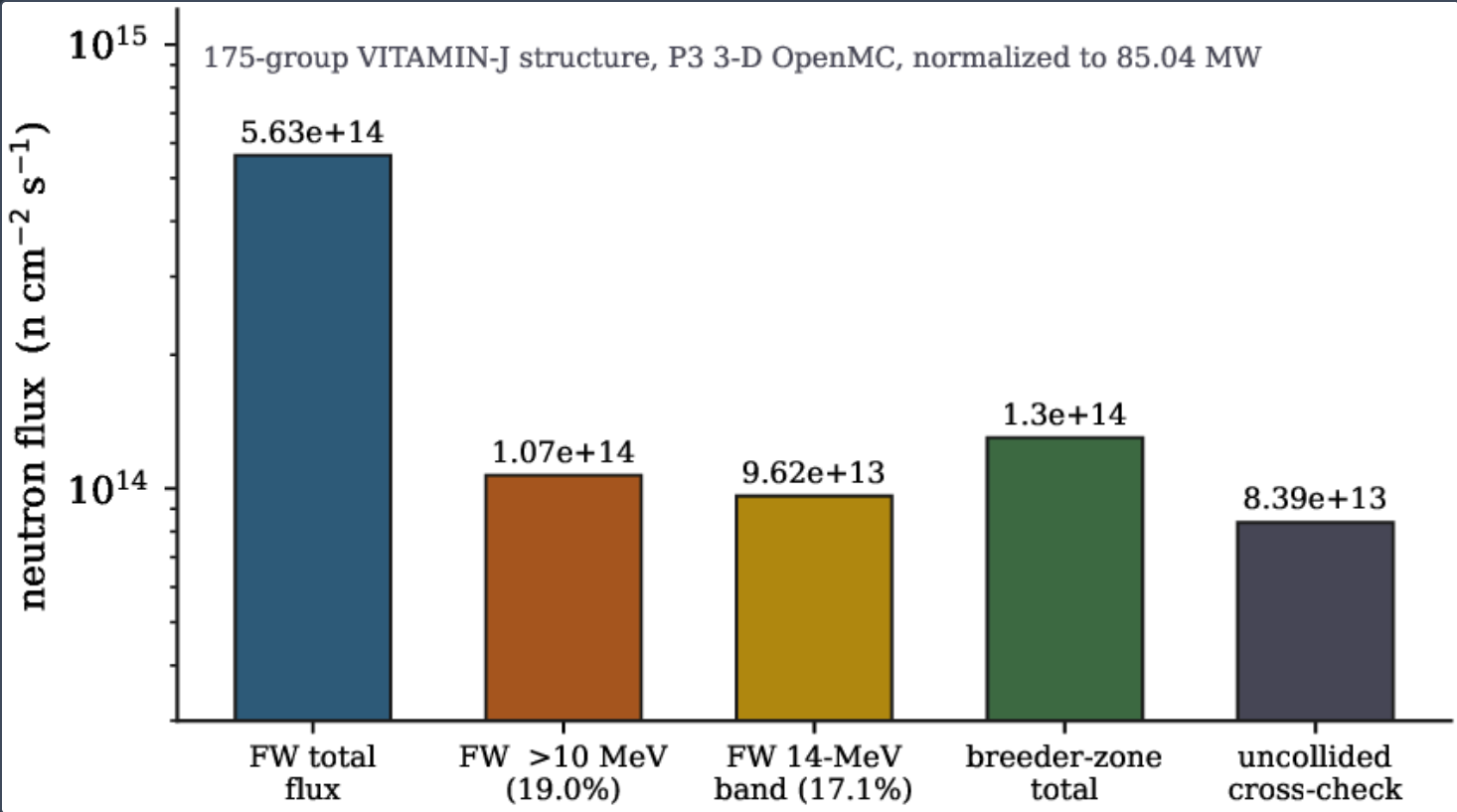
Left: a tungsten penetration plug in the widest (1.0 m) straight end duct returns **0.15×** the open-duct dose, a **6.5×** reduction (penetration-streaming Monte-Carlo study). Right: attenuation of the transmitted field behind the penetration with composite-shield thickness at a reference e-fold length $\lambda \simeq 0.16\text{ m}$; the 40 cm inboard streaming shield marker corresponds to the beam-line dog-leg (BOM-06).

QUANTITY	VALUE	SOURCE
port-area budget	0.719 m ²	BOM-06
port-area demand	0.720 m ²	BOM-06
neutral-beam share of ports	67%	BOM-06
streaming TBR penalty (dog-leg + 40 cm shield)	−0.4% (0.742 → 0.739)	BOM-06
net TBR, 3-D CAD with ports (single-null)	1.12	BV-05B
net TBR, 3-D CAD with ports (double-null)	1.07	BV-05B
tungsten-plug dose factor (1.0 m duct)	0.15× (6.5×)	register
reference attenuation e-fold λ	0.16 m	register

Penetration-streaming and port summary.

FIRST-WALL SPECTRUM AND THE R2S SOURCE TERM

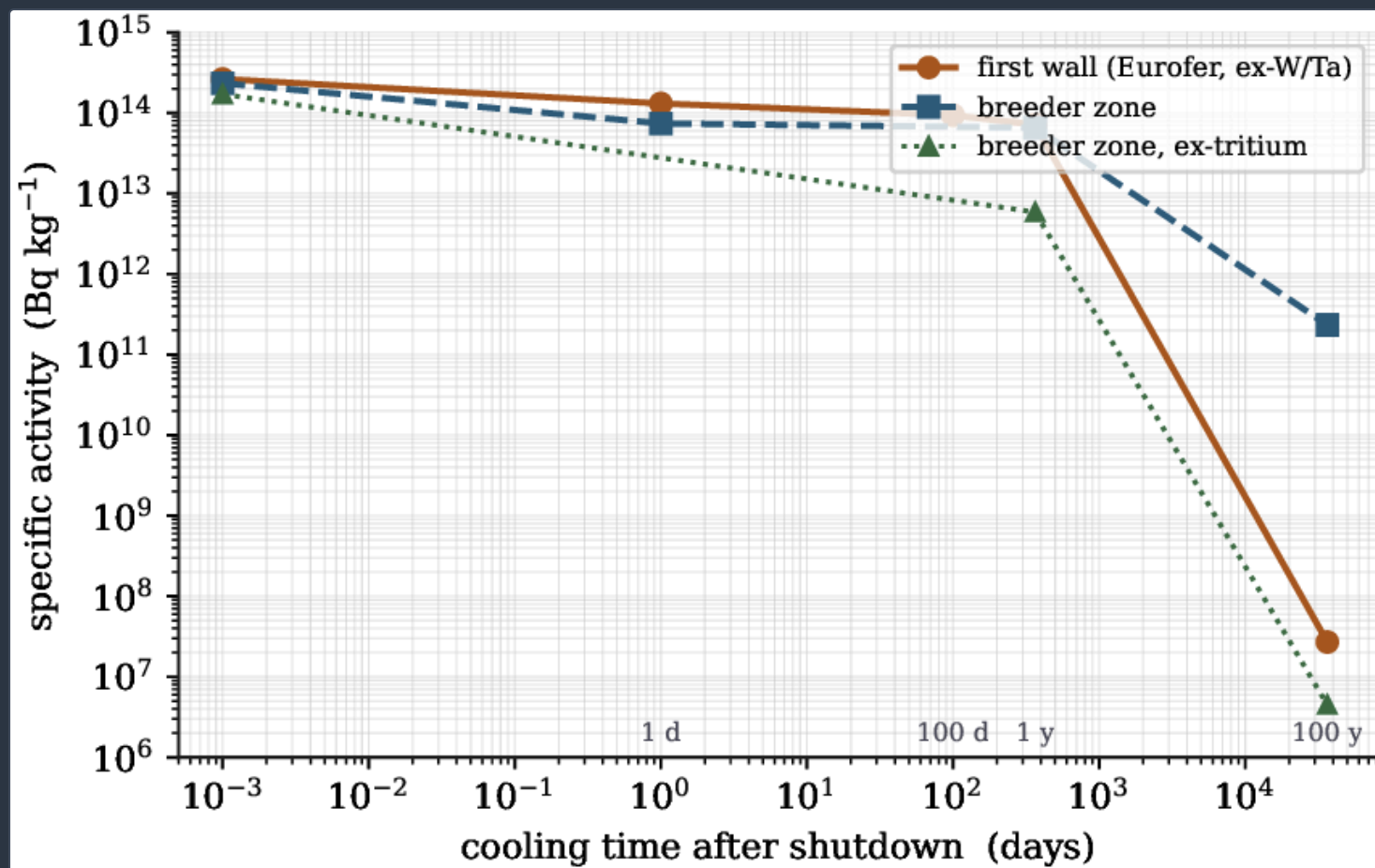
The activation and dose calculation is normalized to the same design power. On the P3 three-dimensional OpenMC model the first-wall total neutron flux is 5.628e14 nps, of which 1.071e14 nps (19.0%) lies above 10 MeV and 17.1% falls in the 14 MeV source band; the breeder-zone flux is 1.302e14 nps (BR-L2-A23). An uncollided-current cross-check at the 1.89 MW/squarem wall load returns 8.390e13 nps, consistent with the tallied first-flight component (Figure 5), which verifies the multigroup condensation 19. These integrals are the spectral source term for the Bateman solve 10.



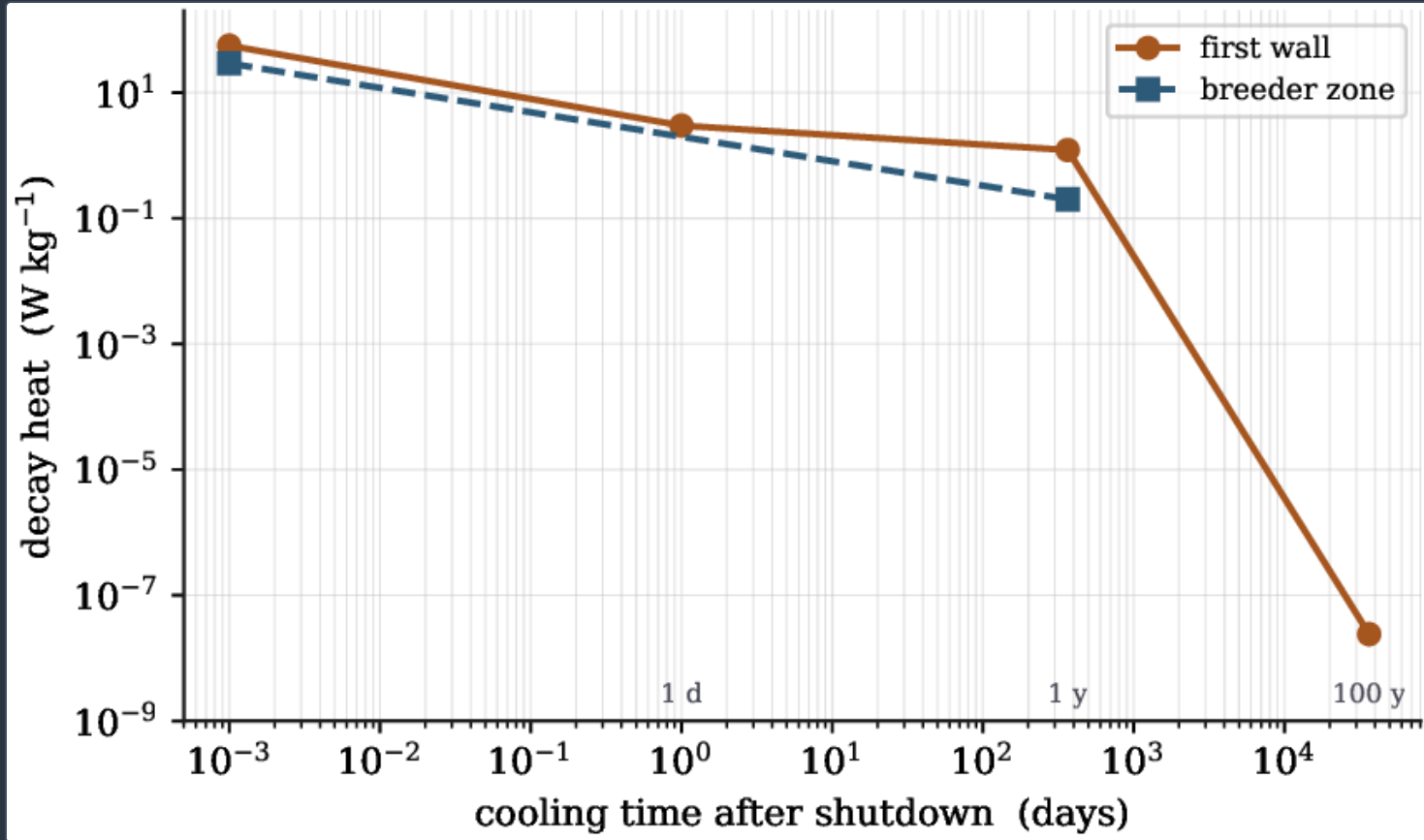
First-wall and breeder-zone spectral partition (175-group VITAMIN-J, P3 three-dimensional OpenMC, normalized to 85.04 MW; BR-L2-A23): first-wall total flux, its **>10 MeV** and 14 MeV-band components, the breeder-zone total, and the uncollided-current cross-check.

ACTIVATION INVENTORY, DECAY HEAT, AND WASTE CLASS

Feeding the group flux to ALARA over a one-full-power-year irradiation and cooling the two-zone slab (3 cm first wall of reduced-activation Eurofer-97, 50 cm breeder zone of Eurofer/beryllium/ Li_4SiO_4 at 90% ^6Li) gives the inventory of Table 3 and Figures 6–8. The first-wall specific activity 12 is 2.64×10^{14} becquerel/kg at shutdown, decaying to 1.31×10^{14} at one day, 9.37×10^{13} at 100 days, 7.13×10^{13} at one year and 2.70×10^7 becquerel/kg at 100 years; the shutdown field is dominated by short-lived ^{56}Mn and ^{52}V , the day-to-year field by ^{55}Fe , ^{54}Mn , ^{51}Cr and ^{49}V , and the century residue by ^3H and ^{53}Mn . The first-wall decay heat 12 falls from 56.0 W/kg at shutdown to 2.99 at one day and 1.22 W/kg at one year (Figure 7); the breeder-zone activity and decay heat are comparable at shutdown (2.34×10^{14} becquerel/kg, 29.2 W/kg) and its century-scale residue is entirely bred tritium held in the ceramic — an artifact of no-purge bookkeeping, since in operation tritium is the recovered product, not a waste term.

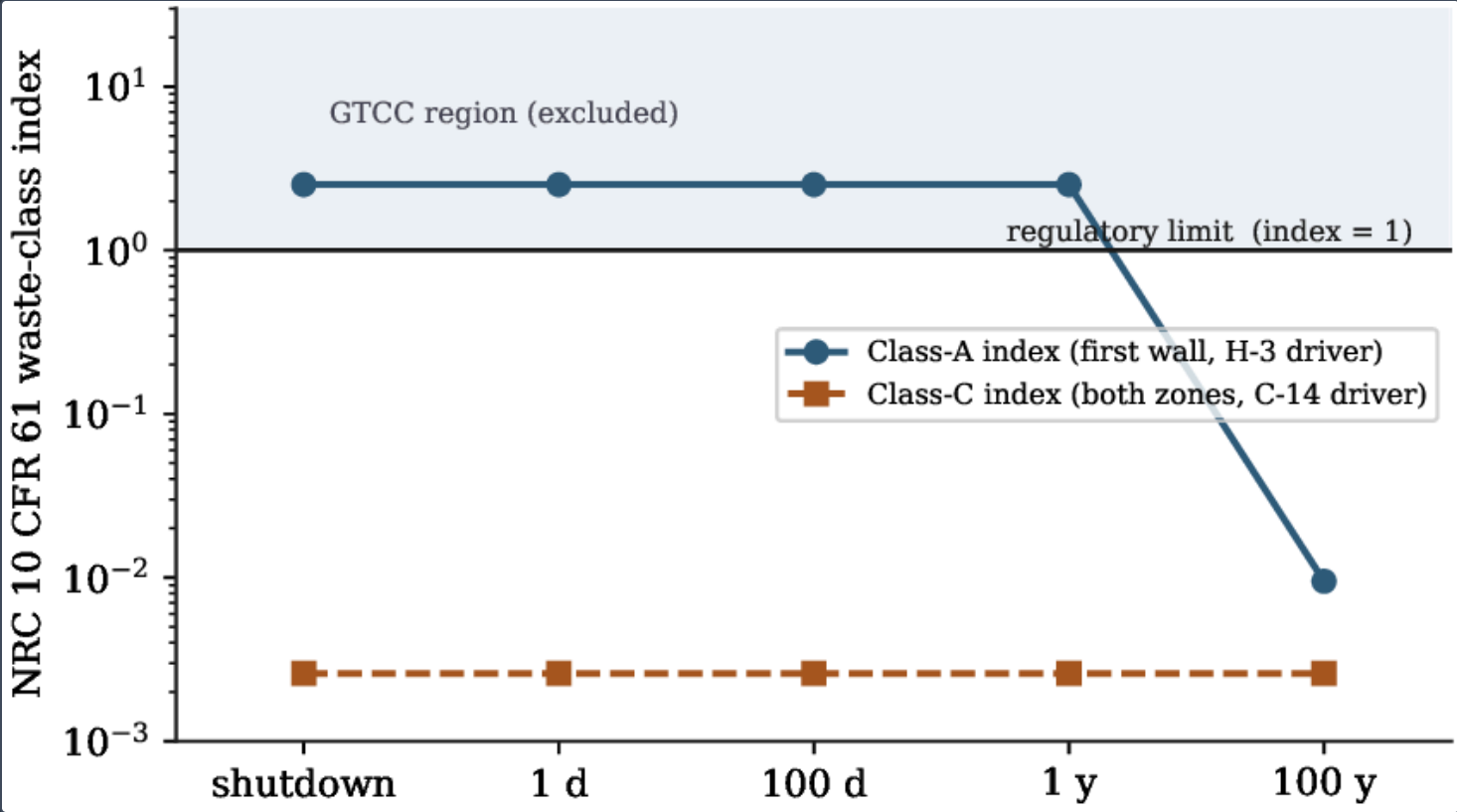


Specific activity versus cooling time for the first wall and breeder zone (BR-L2-A23). The breeder-zone century residue is bred tritium (no-purge bookkeeping); the ex-tritium curve shows the true structural residue.



Decay heat versus cooling time for the first wall and breeder zone (BR-L2-A23). The first-wall decay heat falls below 1.3 W/kg within a year, setting the after-heat basis for the walk-away safety case.

The waste classification follows the NRC 10 CFR 61 concentration-ratio indices computed by ALARA (Figure 8). The Class-C index is $\leq 2.6e-3$ in both zones at all cooling times, driven by ¹⁴C — far below the Class-C boundary, so nothing is greater-than-Class-C and there is no long-lived driver in the modelled composition. The Class-A index is dominated by tritium (first-wall 2.52, falling to 9.5e-3 at 100 years); once tritium is recovered the structure is Class-A low-level waste at discharge. Tritium management, not activation, sets the handling class.



NRC 10 CFR 61 waste-class indices versus cooling time (BR-L2-A23): the Class-C index stays $\leq 2.6e-3$ in both zones at all times (nothing greater-than-Class-C), while the Class-A index is tritium-driven. The structure is low-level waste at discharge.

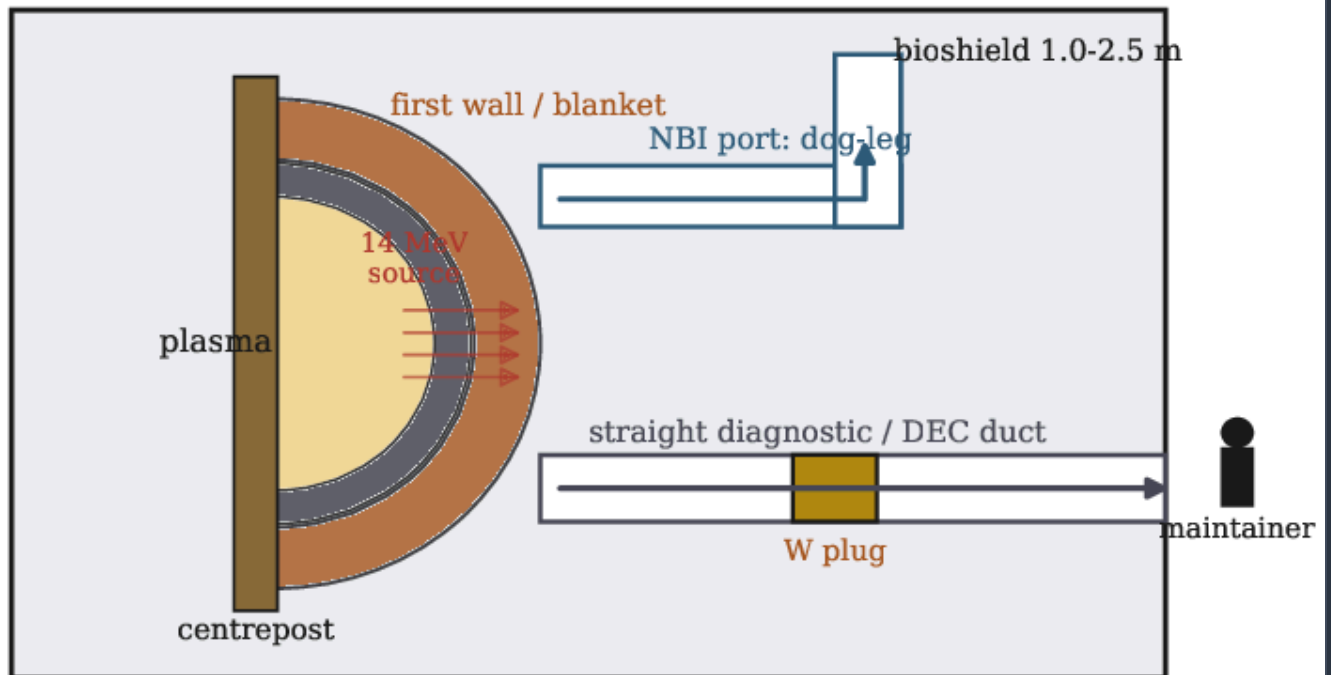
QUANTITY	FIRST WALL	BREEDER ZONE	NOTE
neutron flux (nps)	5.628e14	1.302e14	VITAMIN-J tally
specific activity, shutdown (becquerel/kg)	2.64e14	2.34e14	⁵⁶ Mn/ ⁵² V
specific activity, 1 y (becquerel/kg)	7.13e13	6.55e13	⁵⁵ Fe/ ⁵⁴ Mn
specific activity, 100 y (becquerel/kg)	2.70e7	2.33e11	³ H/ ⁵³ Mn
decay heat, shutdown (W/kg)	56.0	29.2	
decay heat, 1 y (W/kg)	1.22	0.20	
10 CFR 61 Class-C index	$\leq 2.6e-3$	$\leq 2.6e-3$	¹⁴ C driver
waste class at discharge	2cLLW (not GTCC)	tritium-managed	

R2S activation inventory and waste class (BR-L2-A23; 1 full-power-year irradiation, 175-group VITAMIN-J, normalized to \$I85.04

THE RECEPTOR: FROM STREAMING SOURCE TO OCCUPATIONAL DOSE

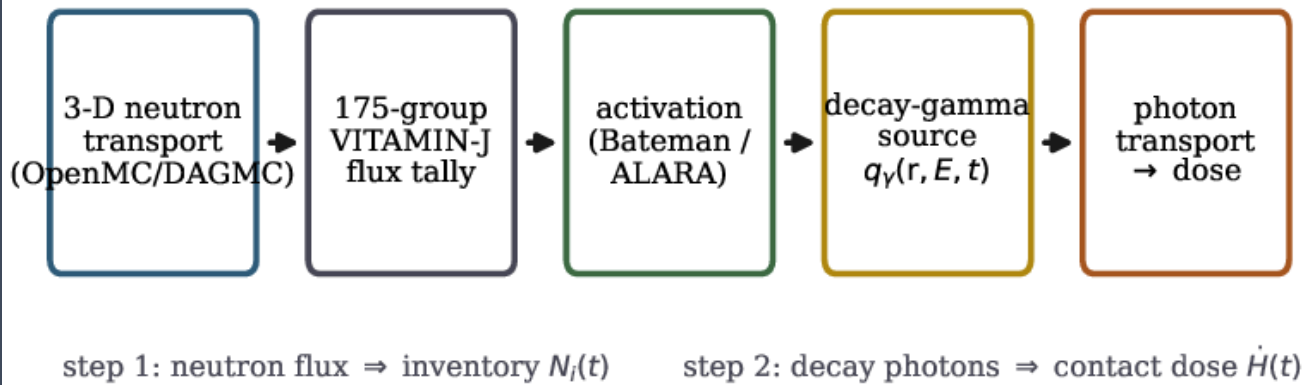
The two halves of the account meet at the maintainer's location (Figure 9). The R2S decay-gamma source 13 is strongest in the activated first wall and blanket, but the field that actually reaches a maintenance position is filtered by the same penetration geometry that governed the neutron leakage: a receptor at the mouth of a bent, plugged and shielded duct sees the wall-albedo tail of 8 and 9, not the line-of-sight source. Because the neutron transport that built the inventory and the photon transport that delivers the dose share one CAD-faithful geometry, the occupational-dose estimate inherits the streaming mitigations directly rather than requiring a separate, decoupled shielding sum (Figure 10). This coherence — one geometry, one transport, from the streaming penetration through the activation to the receptor — is what makes the neutron-safety case auditable.

(Schematic.) Penetration pathways and the shutdown-dose receptor



(Schematic.) Penetration pathways around the breeder: the beam-line dog-leg that breaks the neutron line of sight, the straight diagnostic/direct-conversion duct closed by a tungsten plug, the inboard streaming shield, and the bioshield behind which the maintainer stands. Not to scale.

(Schematic.) Rigorous-two-step shutdown-dose chain on one CAD-faithful geometry



(Schematic.) The rigorous-two-step chain of Section 2: neutron transport fixes the 175-group flux, the Bateman/ALARA solve builds the activation inventory, and the decay-gamma source is transported back through the *same* CAD-faithful geometry to the receptor dose.

Discussion

QUANTITATIVE COMPARISON WITH PUBLISHED PRACTICE

The streaming result sits squarely within established fusion-neutronics practice while being specific to a compact, thin-inboard-shield spherical tokamak. The α^2/L^2 line-of-sight law and the α^{-2} dog-leg suppression are the textbook results of duct-streaming theory ^[25,26,27], and the tungsten-plug factor of $0.15 \times (\sum_i^P t_p \approx 1.9$ mean free paths) is consistent with the high- Z plug attenuation used in ITER port and diagnostic-penetration shielding studies ^[15,14]. The need for adjoint-driven variance reduction to make the deep-streaming FOM 16 tolerable, and the choice of CADIS/FW-CADIS weight windows, follow the same reasoning that governs ITER-scale Monte-Carlo shielding ^[10,11,13]. Where the compact ST differs is in the *margin*: the port budget saturates at 0.719 m² with zero headroom (BOM-06), so the breeding penalty of penetration is not a small perturbation to be absorbed but a first-order design constraint — which is exactly why the with-ports TBR of **1.12/1.07** (BV-05B), still above the 1.05 floor, is the load-bearing breeding number rather than the idealized **1.17**.

THE DOSE CASE IN CONTEXT

On the activation side, the first-wall specific activity of 2.64e14 becquerel/kg at shutdown and the decay heat of 56.0 W/kg are typical of a reduced-activation ferritic–martensitic first wall under 14 MeV loading ^[19,20], and the collapse of the Class-C index to $\leq 2.6e - 3$ places the structure firmly in the low-level-waste class, consistent with the fusion activated-material management literature ^[22]. The result that tritium management — not structural activation — sets the handling class is the same conclusion reached for helium-cooled ceramic-breeder blankets ^[21], and it is what allows the wider Kronos platform to argue a byproduct-material regulatory posture rather than a power-reactor one ^[38,37]. The shutdown-dose model reported here is the validated neutron-and-photon source term on which that occupational-safety and decommissioning case is built, and it connects directly to the neutronics-and-activation digital twin ^[33], the component-activation and waste-routing study ^[34], and the centrepost fluence-and-shielding life-limiter ^[35] that share this transport basis. The nuclear-data uncertainty that dominates the breeding error bar is treated separately ^[36].

Limitations

One honest gate remains. The R2S model resolves the activation-gamma source on the CAD-faithful geometry and the streaming factors on the analytic-plus-Monte-Carlo basis of Sections 2–5, but the *location-specific* occupational dose *rate* at each named maintenance position — the final $\dot{H}(\mathbf{r}_0, t_c)$ of 14 at every workstation — is the staged product of the same DAGMC/OpenMC photon-transport step on the built layout, not a new toolchain. It is computed on the identical geometry and source term reported here; what remains is to enumerate the receptor positions of the final maintenance plan and tabulate the per-position rates. No new physics or code is required, only the frozen layout.

Conclusion

The Hyperion breeder's third-leg neutron account — the streaming penetrations, the breeding penalty they impose, and the occupational dose they set — is closed in three dimensions on CAD-faithful geometry. A $3.009\text{e}19$ /s, 67.8 MW source drives a transport with a **0.121** leakage fraction, a $7.0\text{e}14$ nps first-wall flux and 74.6 MW of nuclear heating. The line-of-sight α^2/L^2 streaming law and its α^{-2} dog-leg suppression, derived here and confirmed by three-dimensional Monte-Carlo, explain why the port-saturated (**0.719**/0.720 m²) layout holds its breeding penalty to **−0.4%** with a beam-line dog-leg and a 40 cm shield, and why a tungsten penetration plug (**6.5×** reduction) is the load-bearing mitigator of the one duct that cannot be bent. With those mitigations the net TBR with real ports settles at **1.12/1.07**, above the 1.05 floor. The rigorous-two-step model, built on the same geometry, gives a first-wall shutdown activity of $2.64\text{e}14$ becquerel/kg decaying to $7.13\text{e}13$ becquerel/kg at one year, a decay heat of **56.0** → **1.22 W/kg**, and a Class-C index $\leq 2.6e - 3$ in both zones at all times — low-level waste, tritium-limited, nothing greater-than-Class-C. The account is complete, CAD-faithful and auditable end to end, from the streaming penetration to the maintainer's dose.

Acknowledgments Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos de-risking campaign for the frozen neutronics, streaming and activation serials on which this work is grounded.

Reproducibility. The source term, transport balance, CAD-faithful chain, port budget and shutdown-dose spectra are frozen results in the Kronos Physics De-Risking Register (BR-OUT-3, BR-L1-A5, BR-L2-A22, BR-L2-A7b, BV-05B, BOM-06, BR-L2-A23, KX-L1-A3), regenerable from the archived OpenMC/DAGMC/ALARA decks with fixed seeds and recorded nuclear-data revisions. This paper is archived at [doi:10.5281/zenodo.22645842](https://doi.org/10.5281/zenodo.22645842); official version: <https://www.kronosfusionenergy.com/publications/neutron-streaming-duct-and-port-penetrations-and-occupa.html> and the register at [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057); the figure-generation script that renders every data figure from the frozen numbers is deposited with this paper.

Data availability The neutron-balance, port-streaming, spectral and activation tables are archived with the deposit ([doi:10.5281/zenodo.22645842](https://doi.org/10.5281/zenodo.22645842)) and reference the Kronos 2026 series. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests Provisional patent applications relating to aspects of this work have been filed.

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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